

Task 1: Session 1 (AIML) - Assignment Questions

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Domain: AIML

College & Branch Name: Zeal Polytechnic, Diploma In AIML

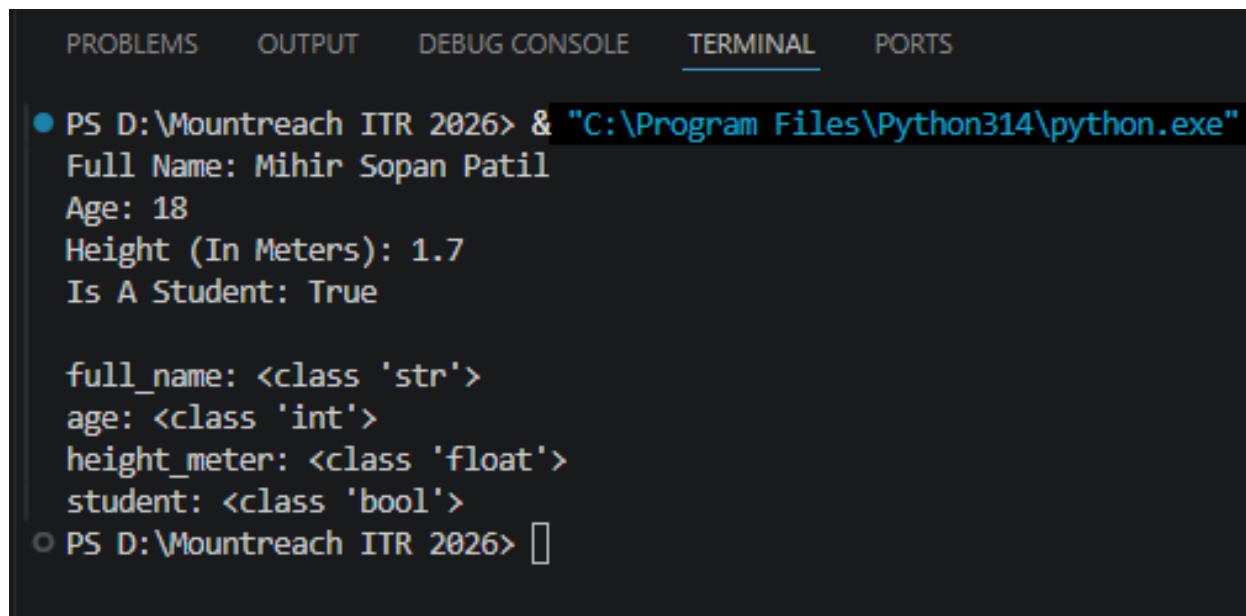
1. Variables & Naming Rules

Create variables to store the following information about yourself:

- Your full name (string)
- Your age (integer)
- Your height in meters (float)
- Whether you are a student (boolean)

Print all four variables. Then explain in comments why you chose those variable names and check their data types using `type()`.

Output:



```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

● PS D:\Mountreach ITR 2026> & "C:\Program Files\Python314\python.exe"
Full Name: Mihir Sopan Patil
Age: 18
Height (In Meters): 1.7
Is A Student: True

full_name: <class 'str'>
age: <class 'int'>
height_meter: <class 'float'>
student: <class 'bool'>
○ PS D:\Mountreach ITR 2026> □
```

Program Code:

```
# Storing Information In Variables
full_name = "Mihir Sopan Patil"    # stores a person's full name(String)
age = 18                          # stores age in whole numbers(Integer)
height_meter = 1.7                 # stores height in meters(Float)
student = True                     # stores student True/False
information(Boolean)

# Printing The Data Stored In Variables
print("Full Name:", full_name)
print("Age:", age)
print("Height (In Meters):", height_meter)
print("Is A Student:", student)

# Checking Data Types Of Variables
print("\nfull_name:", type(full_name))
print("age:", type(age))
print("height_meter:", type(height_meter))
print("student:", type(student))
```

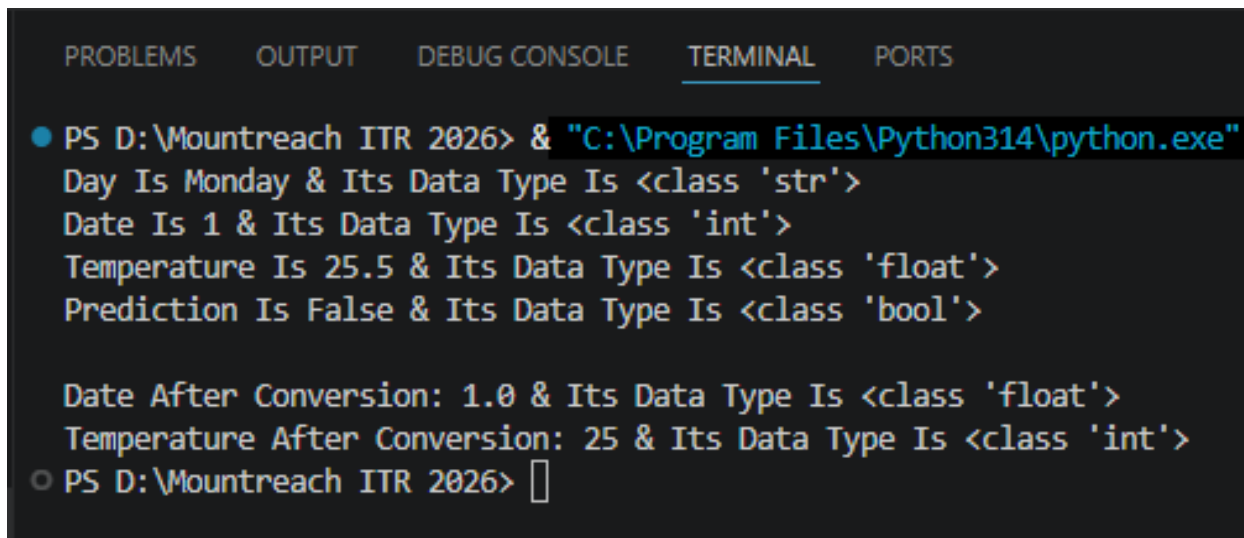
2. Data Types

Write a Python program that:

- Creates one variable of each core data type: str, int, float, and bool.
- Prints each variable along with its data type using type().

Try converting the integer to float and float to integer. Print the results and explain the output.

Output:



```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
● PS D:\Mountreach ITR 2026> & "C:\Program Files\Python314\python.exe"
Day Is Monday & Its Data Type Is <class 'str'>
Date Is 1 & Its Data Type Is <class 'int'>
Temperature Is 25.5 & Its Data Type Is <class 'float'>
Prediction Is False & Its Data Type Is <class 'bool'>

Date After Conversion: 1.0 & Its Data Type Is <class 'float'>
Temperature After Conversion: 25 & Its Data Type Is <class 'int'>
○ PS D:\Mountreach ITR 2026> █
```

Program Code:

```
# Creates One Variable Of Each Core Data Type
day = "Monday"          # String
date = 1                 # Integer
temperature = 25.5       # Float
prediction = False       # Boolean

# Prints Each Variable Along With Its Data Type Using type()
print("Day Is", day, "& Its Data Type Is", type(day))
print("Date Is", date, "& Its Data Type Is", type(date))
print("Temperature Is", temperature, "& Its Data Type Is", type(temperature))
print("Prediction Is", prediction, "& Its Data Type Is", type(prediction))

# Converting The Integer To Float & Float To Integer
date= float(date)        # Converts Integer To Float
temperature= int(temperature) # Converts Float To Integer

# Print The Results
print("\nDate After Conversion:", date, "& Its Data Type Is", type(date))
```

```
print("Temperature After Conversion:", temperature, "& Its Data Type Is",  
type(temperature))  
  
# After Converting The Integer To Float, The Value Of Date Becomes 1.0  
# After Converting The Float To Integer, The Value Of Temperature Becomes 25  
(The Decimal Part Is Removed)
```

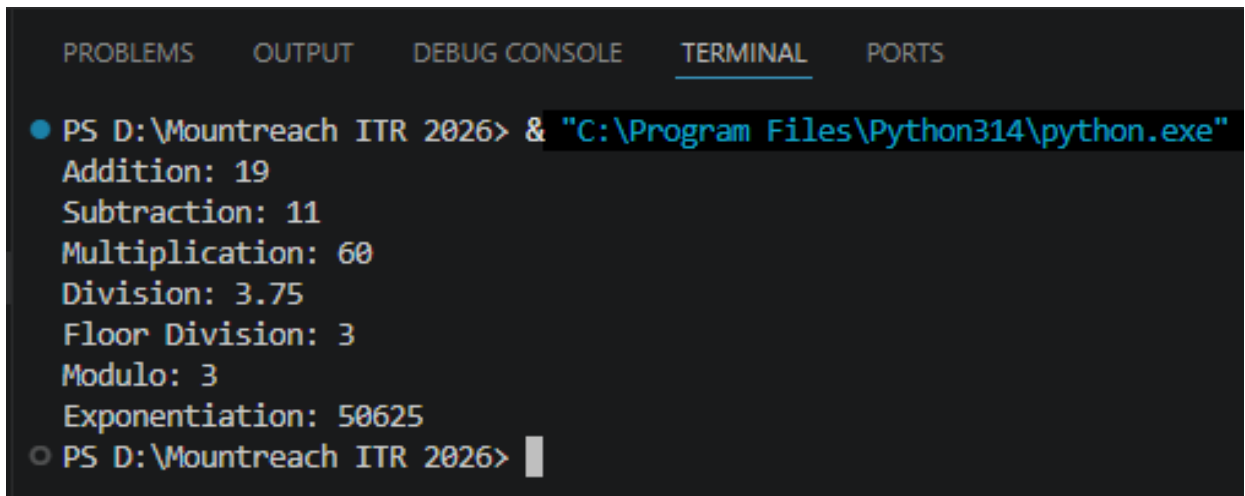
3. Operators - Arithmetic

Write a program that takes two numbers $a = 15$ and $b = 4$ and prints the results of:

- Addition, Subtraction, Multiplication, Division, Floor Division, Modulo, and Exponentiation.

Add comments explaining the difference between `/` and `//` operators with an example.

Output:



```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
● PS D:\Mountreach ITR 2026> & "C:\Program Files\Python314\python.exe"
Addition: 19
Subtraction: 11
Multiplication: 60
Division: 3.75
Floor Division: 3
Modulo: 3
Exponentiation: 50625
○ PS D:\Mountreach ITR 2026> |
```

Program Code:

```
# Arithmetic Operators In Python
a = 15
b = 4

# Performing Arithmetic Operations
print("Addition:", a + b)      # +
print("Subtraction:", a - b)   # -
print("Multiplication:", a * b) # *
print("Division:", a / b)      # /
print("Floor Division:", a // b) # //
print("Modulo:", a % b)        # %
print("Exponentiation:", a ** b) # **

# / (Division) Returns The Exact Quotient As A Floating-Point Number.
# // (Floor Division) Returns The Quotient Rounded Down To The Nearest Whole Number.
```

4. Operators - Comparison & Logical

Given:

```
x = 25
```

```
y = 30
```

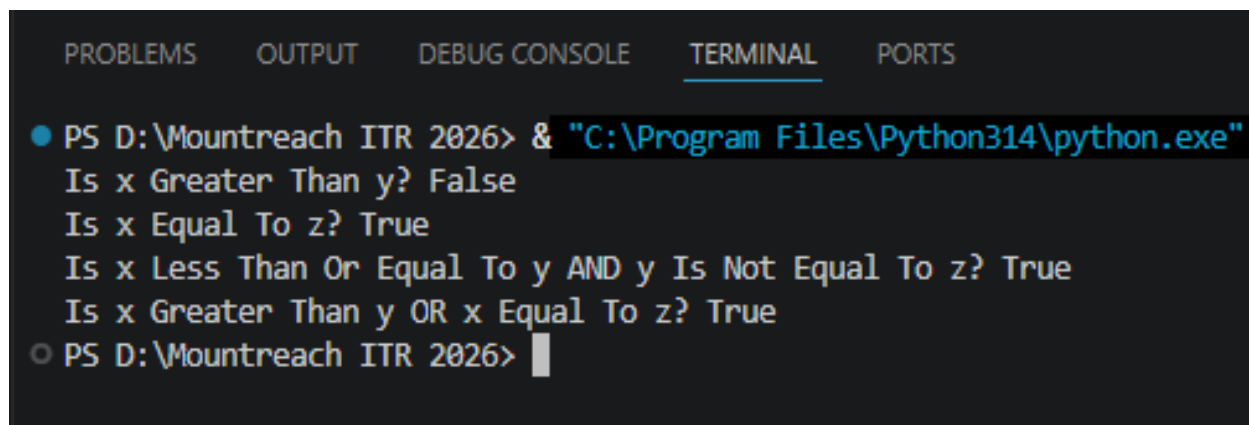
```
z = 25
```

Write expressions using comparison and logical operators to check:

- Is x greater than y?
- Is x equal to z?
- Is x less than or equal to y AND y is not equal to z?
- Is x greater than y OR x equal to z?

Print all results with descriptive messages.

Output:



```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
● PS D:\Mountreach ITR 2026> & "C:\Program Files\Python314\python.exe"
Is x Greater Than y? False
Is x Equal To z? True
Is x Less Than Or Equal To y AND y Is Not Equal To z? True
Is x Greater Than y OR x Equal To z? True
○ PS D:\Mountreach ITR 2026> 
```

Program Code:

```
# Given Values
x = 25
y = 30
z = 25

# Is x Greater Than y?
print("Is x Greater Than y?", x > y) # False
# Is x Equal To z?
print("Is x Equal To z?", x == z) # True
```

```
# Is x Less Than Or Equal To y AND y Is Not Equal To z?
print("Is x Less Than Or Equal To y AND y Is Not Equal To z?", x <= y and y !=
z) # True
# Is x Greater Than y OR x Equal To z?
print("Is x Greater Than y OR x Equal To z?", x > y or x == z) # True
```

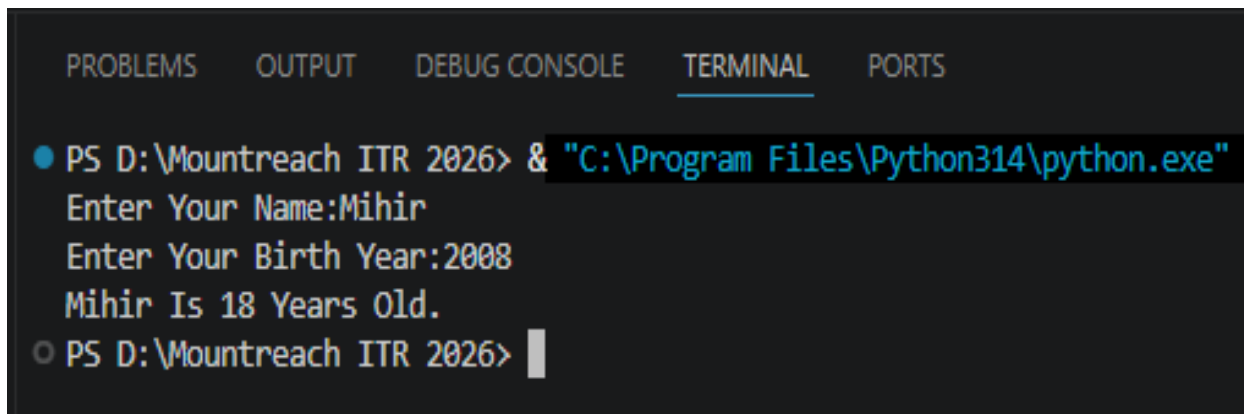
5. Input/Output

Write a program that asks the user to input:

- Their name
- Their birth year

Then calculate and print their current age (assume current year is 2026). Use proper formatting and add comments explaining why you need to convert the input.

Output:



```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
● PS D:\Mountreach ITR 2026> & "C:\Program Files\Python314\python.exe"
Enter Your Name:Mihir
Enter Your Birth Year:2008
Mihir Is 18 Years Old.
○ PS D:\Mountreach ITR 2026> 
```

Program Code:

```
# Input From User
name = input("Enter Your Name:")
birth_year = int(input("Enter Your Birth Year:"))

#input() Always Returns String, Even If The User Types A Number
#To Do Arithmetic Operations, We Convert The String To An Integer Using int()

# Assume The Current Year Is 2026
current_year = 2026

# Calculate Age
age = current_year - birth_year

# Print The Result With Proper Formatting
print(name,"Is",age,"Years Old.")
```

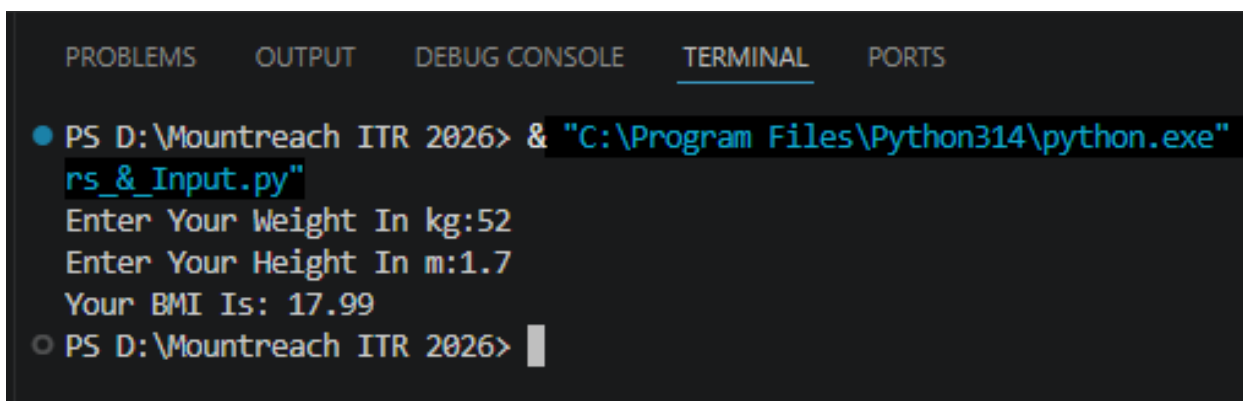

6. Combined - Variables, Operators & Input

Create a simple **BMI Calculator**:

- Take height (in meters) and weight (in kg) as input from the user.
- Calculate BMI using the formula: $BMI = weight / (height ** 2)$
- Print the BMI rounded to 2 decimal places.

Include appropriate comments.

Output:

A screenshot of a terminal window with a dark background. At the top, there are tabs labeled 'PROBLEMS', 'OUTPUT', 'DEBUG CONSOLE', 'TERMINAL' (which is selected and underlined), and 'PORTS'. The terminal shows a command prompt 'PS D:\Mountreach ITR 2026>' followed by the command '& "C:\Program Files\Python314\python.exe" rs_&_Input.py'. The program then prompts 'Enter Your Weight In kg:' with the input '52', and 'Enter Your Height In m:' with the input '1.7'. It then outputs 'Your BMI Is: 17.99'. The prompt returns to 'PS D:\Mountreach ITR 2026>' with a cursor.

Program Code:

```
# BMI Calculator

# Take Weight & Height As Input From The User
weight = float(input("Enter Your Weight In kg:"))
height = float(input("Enter Your Height In m:"))

# Calculate BMI
BMI = weight / (height ** 2)

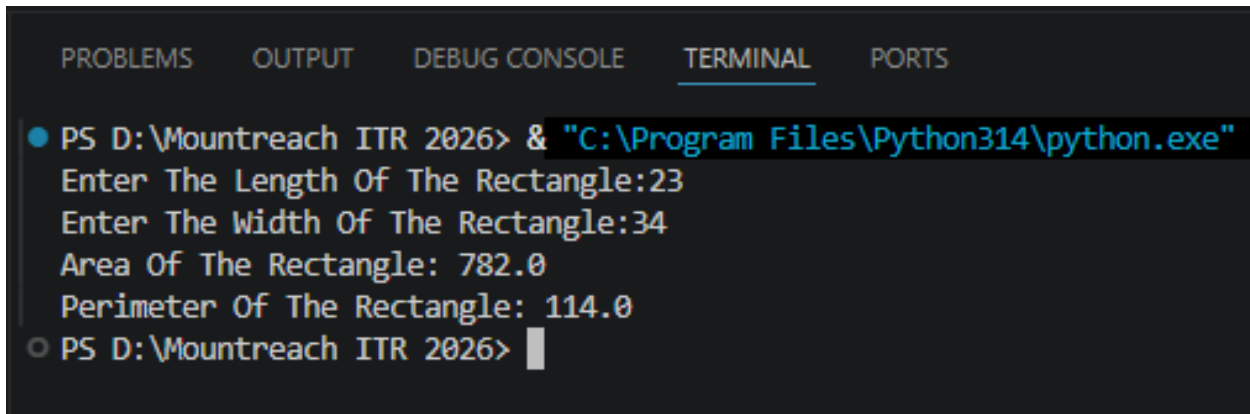
# Print The Result
print("Your BMI Is:", round(BMI, 2))
```

7. Comments

Write a small program that calculates the area and perimeter of a rectangle.

- Take length and width as input.
- Use **at least 5 meaningful comments** (single-line and one multi-line comment).
- Follow good commenting practices (explain why, not what for obvious lines).

Output:



```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
● PS D:\Mountreach ITR 2026> & "C:\Program Files\Python314\python.exe"
Enter The Length Of The Rectangle:23
Enter The Width Of The Rectangle:34
Area Of The Rectangle: 782.0
Perimeter Of The Rectangle: 114.0
○ PS D:\Mountreach ITR 2026> |
```

Program Code:

```
"""
This Program Calculates The Area & Perimeter Of A Rectangle.
The Results Help Us Understand The Size Of The Rectangle And The
Distance Around Its Boundary.
"""

# Taking Input As Float Allows Users To Enter Decimal Values If Needed
length = float(input("Enter The Length Of The Rectangle:"))
width = float(input("Enter The Width Of The Rectangle:"))

# Area Is Useful For Measuring The Space Inside The Rectangle
area = length * width

# Perimeter Is Useful For Finding The Total Distance Around The Rectangle
perimeter = 2 * (length + width)

# Displaying The Results In A Clear And User-Friendly Format
print("Area Of The Rectangle:", area)
```

```
# Showing The Perimeter Separately Makes The Output Easier To Read  
print("Perimeter Of The Rectangle:", perimeter)
```

8. Operators - Assignment & Membership

Given:

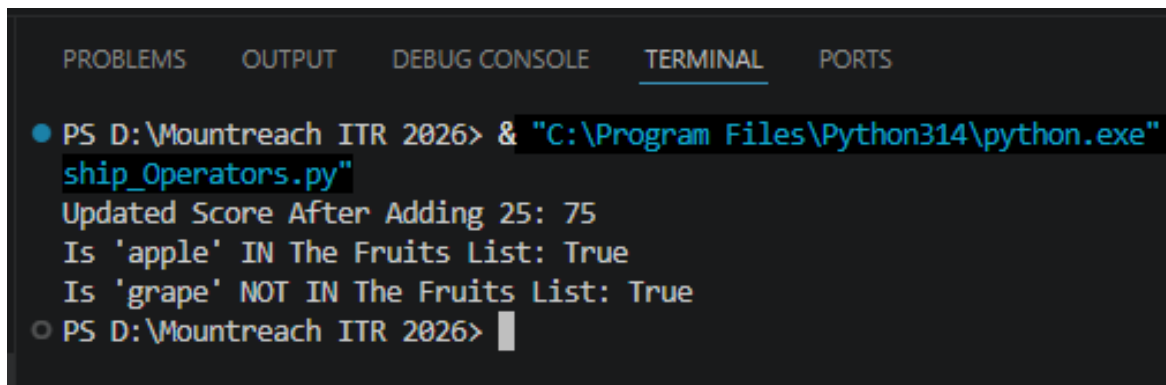
```
fruits = ["mango", "apple", "banana", "cherry"]  
score = 50
```

Perform the following operations:

- Add 25 to score using assignment operator (+=).
- Check if "apple" is in the fruits list.
- Check if "grape" is not in the list.

Print all results with clear messages.

Output:



```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS  
● PS D:\Mountreach ITR 2026> & "C:\Program Files\Python314\python.exe"  
  ship_Operators.py  
  Updated Score After Adding 25: 75  
  Is 'apple' IN The Fruits List: True  
  Is 'grape' NOT IN The Fruits List: True  
○ PS D:\Mountreach ITR 2026> █
```

Program Code:

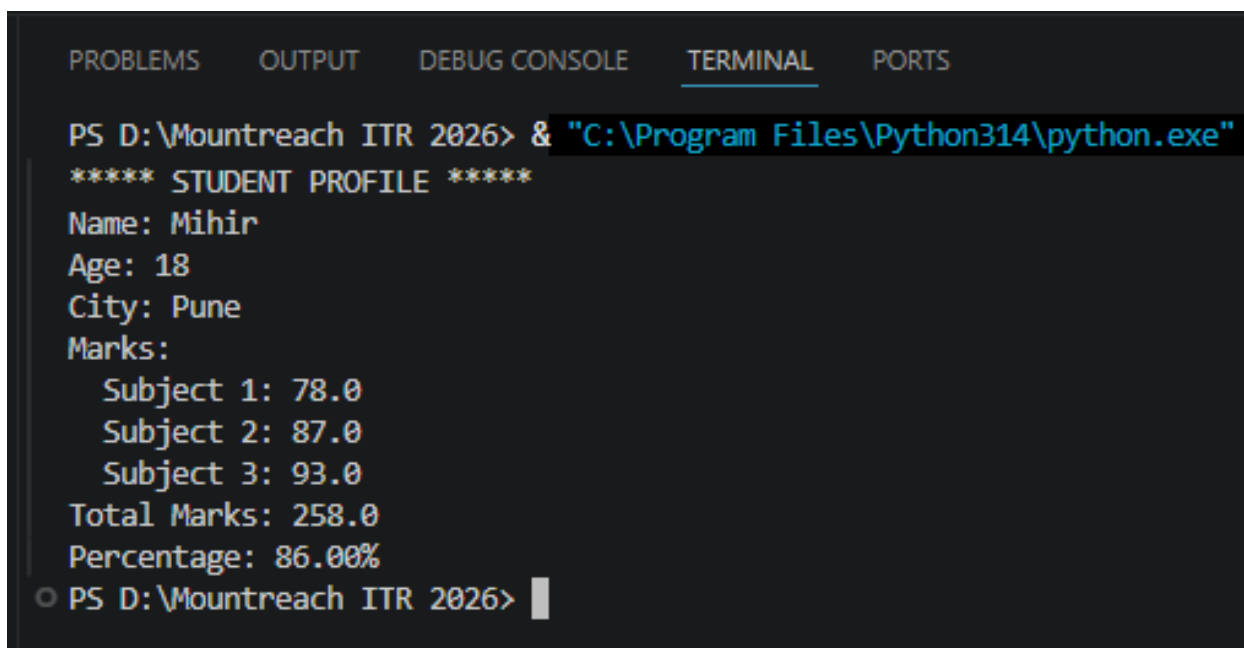
```
# Given List & Variable  
fruits = ["mango", "apple", "banana", "cherry"]  
score = 50  
  
# Add 25 To Score Using The += Assignment Operator  
score += 25 # score = score + 25  
print("Updated Score After Adding 25:", score)  
  
# Check If "apple" Is IN The Fruits List  
has_apple = "apple" in fruits  
print("Is 'apple' IN The Fruits List:", has_apple)  
  
# Check If "grape" Is NOT IN The Fruits List  
no_grape = "grape" not in fruits  
print("Is 'grape' NOT IN The Fruits List:", no_grape)
```

9. Mini Project - All Concepts

Write a Student Profile Generator program that:

- Takes input: name, age, city, and marks in 3 subjects.
- Calculates total marks and percentage.
- Prints a formatted profile with all information.
- Use meaningful variable names, proper comments, and different data types

Output:



```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

PS D:\Mountreach ITR 2026> & "C:\Program Files\Python314\python.exe"
***** STUDENT PROFILE *****
Name: Mihir
Age: 18
City: Pune
Marks:
  Subject 1: 78.0
  Subject 2: 87.0
  Subject 3: 93.0
Total Marks: 258.0
Percentage: 86.00%
PS D:\Mountreach ITR 2026> |
```

Program Code:

```
# Taking Input From The User For Student Profile
name = input("Enter Your Name:")
age = int(input("Enter Your Age:"))
city = input("Enter Your City:")

# Taking Marks For 3 Subjects
marks_subject1 = float(input("Enter Marks For Subject 1:"))
marks_subject2 = float(input("Enter Marks For Subject 2:"))
marks_subject3 = float(input("Enter Marks For Subject 3:"))

# Calculating Total Marks
total_marks = marks_subject1 + marks_subject2 + marks_subject3
```

```
# Calculating Percentage
percentage = (total_marks / 300) * 100 # Assuming Each Subject Is Of 100 Marks

# Printing The Student Profile
print("\n***** STUDENT PROFILE *****")
print("Name:", name)
print("Age:", age)
print("City:", city)
print("Marks:")
print("  Subject 1:", marks_subject1)
print("  Subject 2:", marks_subject2)
print("  Subject 3:", marks_subject3)
print("Total Marks:", total_marks)
print("Percentage:", f"{percentage:.2f}%")
```

10. Debugging & Understanding

The following code has some issues. Correct it and add proper comments explaining the fixes:

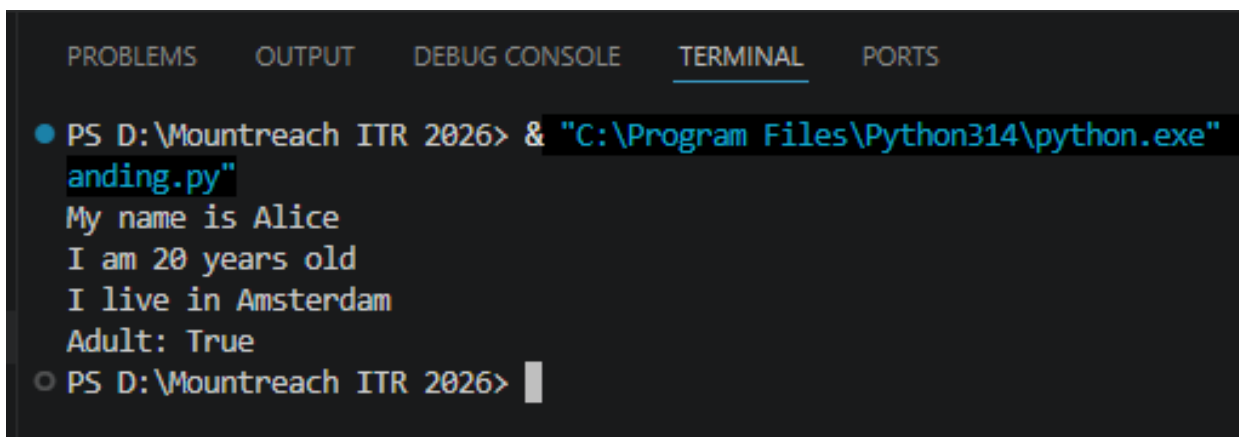
```
Name = "Alice"
age 20
city = Amsterdam

print("My name is" + Name)
print("I am" age "years old")
print("I live in " city)

# Check if age is greater than 18 and print message

print("Adult:" age > 18)
```

Output:



```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
● PS D:\Mountreach ITR 2026> & "C:\Program Files\Python314\python.exe"
  anding.py
  My name is Alice
  I am 20 years old
  I live in Amsterdam
  Adult: True
○ PS D:\Mountreach ITR 2026> |
```

Program Code:

```
Name = "Alice"
age = 20 # Added Assignment Operator (=)
city = "Amsterdam" # Added Quotes Around Text

print("My name is " + Name) # Using + To Join Strings
print("I am", age, "years old") # Using Commas To Separate Values & Variables
print("I live in", city) # Using Commas To Separate Values & Variables

print("Adult:", age > 18) # Using A Comma In Print To Show Result Of Comparison
```